16 The petrol consumption of a car, in litres per 100 kilometres, is given by the formula

$$\text{Petrol consumption} = \frac{100 \times \text{Number of litres of petrol used}}{\text{Number of kilometres travelled}}$$

Nathan's car travelled 148 kilometres, correct to 3 significant figures.
The car used 11.8 litres of petrol, correct to 3 significant figures.

Nathan says,

"My car used less than 8 litres of petrol per 100 kilometres."

Could Nathan be wrong?
You must show how you get your answer.

Distance d: $147.5 \leq d < 148.5$ km

1 mark
Correct rounding bounds

Petrol L: $11.75 \leq L < 11.85$ litres

Largest consumption uses most petrol and least distance:

$$\text{Upper bound} = \frac{100 \times 11.85}{147.5} = 8.033898\ldots$$

1 mark
Use the upper petrol bound and the lower distance bound

Yes. Consumption can exceed 8 litres per 100 km.

For example: $100 \times 11.84 \div 147.5 = 8.0271\ldots > 8$.

1 mark
Correct decision, supported by the bound above 8

(Total for Question 16 is 3 marks)